Problem Chosen	University Physics Competition	Team Control Number
A		758

Controlled Descent and Propellant Estimation for a First-Stage Rocket Booster

Summary

We model the descent and vertical landing of a first-stage booster with coupled translation, pitch rotation, aerodynamic forces, and propellant consumption on a nonrotating spherical Earth. Taking the stated 130 t as dry mass and assuming the booster separates while climbing at 15°, the best trajectory we found coasts for 181 s and then makes a single landing burn from 6.2 km altitude. It needs 25.97 t of propellant at separation, 5.77 t of methane and 20.20 t of liquid oxygen. Of this, 20.89 t is burned and the rest includes the 5 t reserve we require at touchdown. The ship is placed 284.12 km downrange. At contact after 213.25 s the booster is upright, descends at 1.51 m/s, and drifts sideways at 0.003 m/s; replays with time steps down to 0.00125 s meet all the touchdown limits. The separation direction, which the problem leaves open, matters most: separating level or at 30° requires 24.32 or 32.46 t. Effective specific impulse between 300 and 360 s gives 28.48 to 23.98 t, and crossflow drag changes the result by less than 0.5 t. The descent reaches 293 kPa dynamic pressure and an estimated 130–250 kW/m^2 stagnation heat flux before the burn; holding the pressure below 200 or 150 kPa with an extra braking burn raises the requirement to about 37.30 or 50.17 t. These figures assume a rigid booster, a stationary ship, fixed engine and drag parameters, and ideal engine control, and they are local optima.

Contents

1 Introduction **3**
1.1 Background . . . 3
1.2 Restatement of the Problem . . . 3

2 Assumptions **3**

3 Notation **5**

4 Model Force Analysis **5**
4.1 Gravity and Standard Atmosphere . . . 5
4.2 Axial and Crossflow Aerodynamics . . . 6
4.3 Reference Propulsion Parameters . . . 6

5 Model Motion Analysis **7**
5.1 Momentum Balance and Pitch Dynamics . . . 7
5.2 Spherical-Earth Correction . . . 8
5.3 Initial Conditions and Contact Geometry . . . 8
5.4 Analytic Baselines and Scale Estimates . . . 9
5.5 Numerical Integration and Local Powered-Braking Check . . . 9

6 Landing Strategy and Propellant Estimation **10**
6.1 Search and Minimum-Propellant Formulation . . . 10
6.2 Nominal Strategy and Propellant Budget . . . 11
6.3 Spherical-Earth Reference Solution . . . 13
6.4 Dependence on the Separation Direction . . . 13

7 Sensitivity and Flight Loads **15**
7.1 Dependence of the Propellant Budget on Assumptions . . . 15
7.2 Loads Along the Nominal Descent . . . 16
7.3 Aerodynamic Heating Estimate . . . 17
7.4 Cost of Limiting Dynamic Pressure . . . 17
7.5 A Limited Guidance Extension . . . 18
7.6 Summary of the Parameter Studies . . . 18

8 Model Evaluation **18**
8.1 Internal Verification and Numerical Convergence . . . 18
8.2 Strengths and Weaknesses . . . 19

9 Discussion and Conclusions **20**
9.1 Attitude Stability and Vertical Touchdown . . . 20
9.2 Loads and Scope . . . 20
9.3 Earth Curvature and Rotation . . . 20
9.4 Ship Placement, Wind, and Deck Motion . . . 20
9.5 Conclusions and Practical Recommendations . . . 20

A Reproducibility and Source Code **22**

Report on the Use of AI **24**

1 Introduction

1.1 Background

A reusable first stage has to come back intact. After separation it is still moving at several times the speed of sound, high in the atmosphere, and it must lose that speed, turn upright, and settle onto a landing site gently enough to be flown again. For a sea landing the recovery ship can be placed under the descent path, so the booster does not have to fly back toward the launch site. What remains is to brake, to keep the attitude under control, and to touch down slowly with some propellant in reserve. The propellant carried for this descent is mass that the first stage cannot use during ascent, so the size of the descent budget matters for the whole vehicle.

This paper estimates that budget. Sections 2–5 set out the assumptions and the force and motion model, including the correction for a spherical Earth. Section 6 finds a minimum-propellant landing strategy, reports the separation inventory, and shows how it depends on the separation direction. Section 7 examines engine efficiency, aerodynamic drag, and the loads along the descent, and the final sections evaluate the model and state the conclusions.

1.2 Restatement of the Problem

At separation the booster is at 70 km altitude and moves at 6000 km/h. It is 65 m tall and 7.0 m in diameter, and the problem figure gives its dry mass as 130000 kg. It burns liquid oxygen and methane. The recovery ship may be placed wherever is convenient. We are asked to describe a safe descent and a controlled vertical landing and to estimate how much fuel the booster must carry at separation [10]. Its total mass at separation is therefore the dry mass plus the propellant on board.

The problem gives the speed at separation but not its direction. The direction sets how long the booster stays aloft, how far it travels downrange, and its state when the landing burn begins. It is fixed by the ascent mission, not by the recovery plan, so we treat it as an input, with 15° above horizontal as the reference (Section 6.4). Throughout, *propellant* means the combined mass of liquid oxygen and methane, *fuel* means the methane alone, and the *inventory* is the propellant mass on board at separation.

2 Assumptions

- **Vehicle.** The booster is a rigid circular cylinder, 65 m long and 7 m in diameter. Its 130 t dry mass is distributed uniformly for the pitch inertia, and the remaining propellant is treated as a point mass at the center of mass (Section 5). Pitch rotation is modeled because the offset between the center of mass and the center of pressure creates aerodynamic moments, so an upright touchdown has to follow from the dynamics. Structural and thermal capacity are outside the study; the loads are reported as requirements.

- **Earth and atmosphere.** The reference calculation uses a nonrotating spherical Earth with inverse-square gravity and the 1976 U.S. Standard Atmosphere, continued isothermally above 84.852 km geopotential altitude where needed. A flat-Earth

version of the same equations provides starting guesses and the earlier planar comparisons. Motion stays in the vertical plane of the separation velocity, and the nominal case has no wind.

- **Separation state.** The booster separates at 70 km altitude and 6000 km/h. The flight-path angle, which the problem does not give, is 15° above horizontal in the reference case and 0° or 30° in the alternatives. The body axis is initially aligned with the velocity and the pitch rate is zero.

- **Propulsion.** One engine provides up to 2.846 MN, throttleable between 40% and 100%, with a ±10° gimbal, an effective specific impulse of 330 s, and an oxygen-to-methane mass ratio of 3.5, so methane is 1/4.5 of the propellant mass. At most three ignitions are allowed. Throttle and gimbal respond instantaneously, with no ignition delay or minimum burn time. There are no fins or reaction-control thrusters, so the attitude evolves passively while the engine is off.

- **Aerodynamics.** Axial and crossflow forces use constant coefficients $C_a = 0.8$ and $C_n = 1.2$, with the center of pressure $0.05L$ ahead of the center of mass.

- **Ship and touchdown.** The ship is stationary, with a level deck at sea level, and may be placed anywhere along the ground track; each scenario gets its own position. The booster has no landing gear. A landing is accepted when, at first contact, the booster is within 5 m of the ship's position, moves less than 1 m/s horizontally and 2 m/s vertically, tilts less than 2° from upright, rotates slower than 1°/s, and retains at least 5 t of usable propellant. Unusable residual propellant is taken as zero.

- **Parameter studies.** We vary the separation angle (0°, 15°, 30°), the effective specific impulse (300, 330, 360 s), and the crossflow coefficient (0.8, 1.0, 1.2, 1.6), one at a time. The axial coefficient, pressure-center offset, thrust, and reserve are not varied.

3 Notation

Symbol	Meaning	Unit
x	Downrange distance in the flat-Earth model	m
s, h	Surface arc distance and radial COM altitude	m
r, R_E	Distance from Earth center, Earth radius	m
v_x, v_h	Local horizontal and radial velocity	m/s
θ, ω	Pitch from local horizontal, local pitch rate	rad, rad/s
$m, m_d, m_{p,0}$	Total, dry, and separation propellant mass	kg
T, u_T, δ	Thrust, throttle fraction, gimbal angle	N, 1, rad
I_{sp}, g_0	Effective specific impulse, standard gravity	s, m/s^2
L, D, R_b	Body length, diameter, radius	m
A_{end}	End (axial reference) area	m^2
C_a, C_n	Axial and crossflow force coefficients	1
Δl	Pressure-center offset from COM	m
ξ	Axial distance of an aerodynamic strip from the COM	m
I_d	Dry pitch moment of inertia	kg m^2
ρ	Air density	kg/m^3
γ_0	Initial flight-path angle	rad
q	Dynamic pressure, $\rho U^2/2$	Pa

4 Model Force Analysis

4.1 Gravity and Standard Atmosphere

We first write the model in a flat downrange–vertical plane; Section 5.2 then adds Earth curvature for the reference calculation. Position refers to the center of mass (COM), and the pitch angle θ is the angle of the body axis, from tail to nose, above the horizontal. Gravity acts at the COM with magnitude

$$g(h) = g_0 \left(\frac{R_E}{R_E + h} \right)^2, \qquad g_0 = 9.80665 \text{ m/s}^2, \quad R_E = 6356766 \text{ m}. \tag{1}$$

At 70 km this gives $g = 9.5942$ m/s^2, about 2.17% below g_0.

Atmospheric properties follow the lower layers of the 1976 standard atmosphere [3]. Geometric altitude h is converted to geopotential altitude $H = R_E h/(R_E + h)$. In a layer with base (H_b, T_b, p_b) and temperature gradient λ,

$$T_a = T_b + \lambda(H - H_b), \tag{2}$$

$$p_a = \begin{cases} p_b (T_a/T_b)^{-g_0/(R_a \lambda)}, & \lambda \neq 0, \\ p_b \exp[-g_0(H - H_b)/(R_a T_b)], & \lambda = 0, \end{cases} \tag{3}$$

$$\rho = \frac{p_a}{R_a T_a}, \qquad R_a = 287.05287 \text{ J/(kg K)}. \tag{4}$$

These expressions follow from hydrostatic balance, $dp_a/dH = -\rho g_0$, and the ideal-gas law. The tabulated layers end at $H = 84.852$ km. Above that altitude we continue the last layer isothermally, which matters only for the steep separation case of Section 6.4.

4.2 Axial and Crossflow Aerodynamics

The cylindrical approximation gives $A_{end} = \pi D^2/4 = 38.4845$ m^2 and side area $DL = 455$ m^2. Define body directions

$$\mathbf{e} = (\cos\theta, \sin\theta), \quad \mathbf{n} = (-\sin\theta, \cos\theta), \quad \mathbf{u} = \mathbf{v} - \mathbf{w}, \quad u_a = \mathbf{u}\cdot\mathbf{e}, \quad u_n = \mathbf{u}\cdot\mathbf{n}. \tag{5}$$

We split the aerodynamic force into axial and crossflow parts instead of applying one coefficient to a changing projected area, because flow across a cylinder behaves differently from flow along it [2]. The constant coefficients are engineering estimates and are not fitted to that report.

The axial force is

$$F_a = -\tfrac{1}{2}\rho C_a A_{end} u_a |u_a|. \tag{6}$$

For an equivalent crossflow strip at signed axial distance ξ from the COM, rotation adds the local normal velocity $\omega\xi$. Thus

$$dF_n = -\tfrac{1}{2}\rho C_n D(u_n + \omega\xi)|u_n + \omega\xi|\, d\xi. \tag{7}$$

With N midpoint strips of length $\Delta\xi = L/N$ and $\xi_i = -L/2 + (i - 1/2)\Delta\xi + \Delta l$,

$$\mathbf{F}_D = F_a\mathbf{e} + \sum_i F_{n,i}\mathbf{n}, \qquad M_a = \sum_i \xi_i F_{n,i}. \tag{8}$$

Shifting the strips by Δl represents the uncertain position of the center of pressure. Density is evaluated at the COM, and its variation along the 65 m body is neglected. The axial force acts along the centerline. Figure 1 defines the forces and signs.

Without rotation, $M_a = \Delta l F_n$, so even a uniform wind produces a moment when $\Delta l \neq 0$. For a centered distribution in pure rotation, direct integration gives

$$M_a = -\frac{\rho C_n D L^4}{64}\omega|\omega|. \tag{9}$$

which always opposes the rotation. More generally, the power delivered by the air-relative forces satisfies

$$\mathbf{F}_D \cdot \mathbf{u} + M_a\omega = -\tfrac{1}{2}\rho C_a A_{end}|u_a|^3 - \tfrac{1}{2}\rho C_n D \sum_i \Delta\xi\, |u_n + \omega\xi_i|^3 \leq 0. \tag{10}$$

The aerodynamic forces therefore always remove energy from the motion relative to the air, which gives a useful check on signs and implementation. In a wind, the energy measured in the ground frame can still increase.

4.3 Reference Propulsion Parameters

We set the maximum thrust to 2.846 MN, the published thrust of one BE-4 methane–oxygen engine [1], and treat it as a representative single landing engine; the problem does not specify the engine. The effective specific impulse of 330 s, the 40% minimum throttle, and the 10° gimbal limit are design assumptions. The oxygen-to-methane mass ratio of 3.5 follows a methane–oxygen mission study [4]; we do not use that study's vacuum specific impulse, because the landing burn takes place low in the atmosphere. The aerodynamic reference values are $C_a = 0.8$, $C_n = 1.2$, and $\Delta l = 0.05L$.

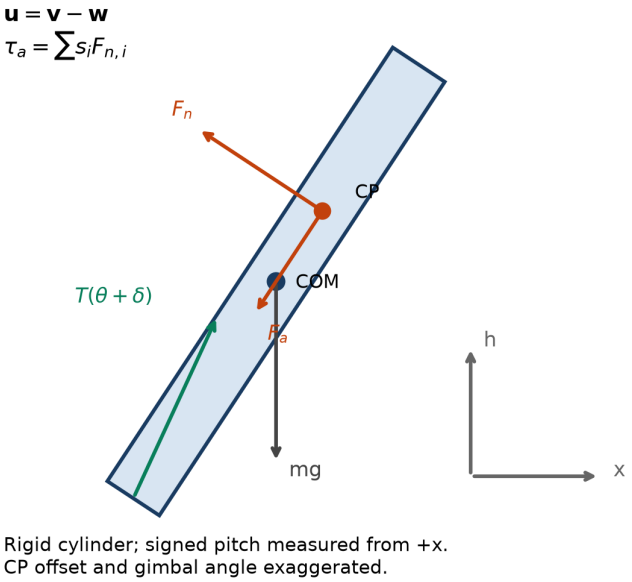


Figure 1: Geometry and force conventions. The pressure-center offset is exaggerated. A positive gimbal angle at the tail produces a negative pitch moment.

The standard effective-exhaust-velocity definition gives $c = I_{sp} g_0$ and

$$\dot{m} = -T/c, \qquad T = 0 \quad \text{or} \quad 0.4T_{max} \le T \le T_{max}. \tag{11}$$

The flow consumes oxygen and methane in the fixed mixture ratio. The variation of thrust with ambient pressure and the engine transients are absorbed into the effective parameters.

5 Model Motion Analysis

5.1 Momentum Balance and Pitch Dynamics

The state is $\mathbf{y} = (x, h, v_x, v_h, m, \theta, \omega)$. Because the thrust already accounts for the momentum carried away by the exhaust, the translational balance is $m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{F}_D + \mathbf{F}_T$, with no separate $\dot{m}\mathbf{v}$ term. In the flat-Earth model the equations are

$$\dot{x} = v_x, \qquad \dot{h} = v_h, \tag{12}$$

$$\dot{v}_x = \frac{F_{D,x} + T\cos(\theta+\delta)}{m}, \qquad \dot{v}_h = -g(h) + \frac{F_{D,h} + T\sin(\theta+\delta)}{m}, \tag{13}$$

$$\dot{m} = -\frac{T}{g_0 I_{sp}}, \qquad \dot{\theta} = \omega, \tag{14}$$

$$I_d\dot{\omega} = M_a - \frac{L}{2}T\sin\delta, \qquad I_d = \frac{m_d(3R_b^2 + L^2)}{12}. \tag{15}$$

Thrust acts at the tail, $-(L/2)\mathbf{e}$, which gives the negative sign in the torque. The dry body is a uniform cylinder, and the remaining propellant is a point mass at the COM, so it adds no pitch inertia and I_d is constant. Propellant spread through real tanks would make the inertia change during the burn and add exhaust angular-momentum terms; both are neglected.

5.2 Spherical-Earth Correction

For the reference calculation we replace the flat geometry with a nonrotating spherical Earth of radius R_E. Write $r = R_E + h$ and $s = R_E\phi$, where s is surface arc distance. Velocities v_x and v_h are the physical components along the local horizontal and radial directions, which turn as the booster moves downrange. The state keeps its seven components, with x replaced by s. Let $a_x = F_x/m$ and $a_h = F_h/m$ denote the non-gravitational accelerations. Differentiating the moving basis gives

$$\dot{s} = \frac{R_E}{r} v_x, \qquad \dot{h} = v_h, \tag{16}$$

$$\dot{v}_x = a_x - \frac{v_x v_h}{r}, \qquad \dot{v}_h = a_h - g(h) + \frac{v_x^2}{r}. \tag{17}$$

The last terms in the velocity equations are the curvature accelerations, and the factor R_E/r converts horizontal speed at altitude into distance along the surface.

We keep $\omega = \dot{\theta}$ as the pitch rate relative to the local horizontal. Because the local horizontal rotates at $-v_x/r$ in the flight plane, the inertial body rate is $\omega_I = \omega - v_x/r$, and the strip velocity in Eq. (8) becomes $u_n + \omega_I \xi_i$. The angular equation is then

$$\dot{\omega} = \frac{M_a + M_T}{I_d} + \frac{a_x}{r} - \frac{2 v_x v_h}{r^2}, \tag{18}$$

where the last two terms come from differentiating v_x/r. With them, a torque-free body keeps a fixed inertial attitude while its pitch measured from the local horizontal changes as it travels downrange.

For contact with the sea-level deck, the longitudinal section of the cylinder is a rectangle in the flight plane. Its point nearest the Earth's center has body coordinates

$$a_* = \mathrm{clip}(-r\sin\theta, -L/2, L/2), \qquad b_* = \mathrm{clip}(-r\cos\theta, -R_b, R_b), \tag{19}$$

and first contact occurs when

$$f_s = \sqrt{(a_*\cos\theta - b_*\sin\theta)^2 + (r + a_*\sin\theta + b_*\cos\theta)^2} - R_E - h_{deck} = 0. \tag{20}$$

The ship is a fixed point on the surface, given by its arc coordinate. Within the 5 m position tolerance the local vertical turns by less than 10^{-6} rad, so the deck at the contact point is level in the local frame.

5.3 Initial Conditions and Contact Geometry

For each selected initial flight-path angle,

$$x_0 = 0, \quad h_0 = 70000 \text{ m}, \quad (v_{x,0}, v_{h,0}) = \frac{6000}{3.6}(\cos\gamma_0, \sin\gamma_0), \quad m_0 = m_d + m_{p,0}. \tag{21}$$

The reference uses $\gamma_0 = \theta_0 = 15^\circ$ and $\omega_0 = 0$. The separation propellant $m_{p,0}$ is an unknown found by the optimization of Section 6.

In the flat-Earth model, a cylinder without landing gear reaches $(L/2)|\sin\theta| + R_b|\cos\theta|$ below its COM, so first contact with the deck occurs when

$$f(\mathbf{y}) = h - h_{deck} - \frac{L}{2}|\sin\theta| - R_b|\cos\theta| = 0. \tag{22}$$

The spherical model uses Eq. (20) instead. An upright booster touches the deck when $h - h_{deck} = L/2$. The contact condition only fixes the moment of touchdown; speed, attitude, rotation rate, reserve, and position relative to the ship are checked at that moment.

5.4 Analytic Baselines and Scale Estimates

For constant gravity, zero thrust, and zero drag,

$$h(t) = h_0 + v_{h,0}t - \tfrac{1}{2}g_0t^2, \qquad x(t) = v_{x,0}t. \tag{23}$$

At 15°, $(v_{x,0}, v_{h,0}) = (1609.88, 431.37)$ m/s. The constant-g_0 apex estimate is 79.49 km after 43.99 s. The integrated spherical trajectory, with drag and variable gravity, reaches 80.1 km. In the variable-gravity vertical vacuum limit, the conserved specific energy is

$$E = \tfrac{1}{2}v_h^2 - \frac{g_0R_E^2}{R_E + h}. \tag{24}$$

Without opposing external acceleration in the thrust direction, integrating $dv = -c\,dm/m$ yields

$$\Delta v = c\ln(m_0/m_f). \tag{25}$$

For 210 t initially and 135 t finally, this gives 1429.86 m/s. This is the ideal velocity change the propellant can provide. The velocity change actually needed differs, because drag removes speed for free while gravity and steering add to the requirement.

With mass frozen, no drag, and vertical maximum thrust, a useful stopping-distance estimate is

$$d_{stop} \simeq \frac{v_h^2}{2(T_{max}/m - g_0)}, \qquad T_{max} > mg_0. \tag{26}$$

For 180 t and a 90 m/s descent it gives 674.5 m, ignoring mass loss and drag. The thrust-to-weight ratio at full thrust grows from 1.382 at 210 t to 2.232 at dry mass, so the changing mass must be part of the dynamics.

5.5 Numerical Integration and Local Powered-Braking Check

A fixed-step fourth-order Runge–Kutta integrator advances the coupled equations. The step shrinks near the deck, and contact is located by linear interpolation across the sign change of Eq. (22), so the event time is less accurate than the integration itself. A propellant guard stops the run before a step would take the mass below the dry mass; it prevents negative propellant but does not find the exact depletion time.

To test the powered and rotational parts of the model on their own, we start a local case at $h = 1200$ m, $v_h = -90$ m/s, $v_x = 0$, and $m = 180000$ kg. Define height above upright contact $z = h - L/2$ and a descent-speed reference

$$v_d = -\min\{90, \sqrt{2k_z\max(z, 0.01) + v_c^2}\}, \quad k_z = 1.5\ \text{m/s}^2, \quad v_c = 1\ \text{m/s}. \tag{27}$$

A vertical acceleration command $a_c = k_v(v_d - v_h)$, $k_v = 4\ \text{s}^{-1}$, supplies the throttle request $[m(g_0 + a_c) - F_{D,h}]/T_{max}$, clipped to the permitted operating range. This test controller acts only along the vertical and uses g_0 for gravity. Pitch control requests

$$M_{req} = I_d[k_p(\pi/2 - \theta) - k_d\omega] - M_a, \quad \delta = \arcsin\left[\text{clip}\left(-\frac{M_{req}}{(L/2)T}, -1, 1\right)\right], \tag{28}$$

where $k_p = 0.5\ \text{s}^{-2}$, $k_d = 1.2\ \text{s}^{-1}$, followed by gimbal clipping. Without saturation this gives the stable error equation $\ddot{e} + k_d\dot{e} + k_pe = 0$. Saturation and engine-off flight break this linear argument, which is why the full trajectory is checked separately.

At a 0.005 s step the nominal case contacts after 35.97505 s with $v_h = -1.50291$ m/s and consumes 22679.12 kg. A separate test with initial pitch 89° and initial rate 0.005 rad/s ends with approximately zero pitch error and $v_x = 0.4671$ m/s. Its horizontal drift is 16.56 m, exceeding the 5 m fixed-ship position tolerance if the ship remains at $x = 0$. The attitude recovers, but the booster misses a fixed ship; reaching the ship is the task of the full-flight optimization in Section 6. Figure 2 shows both runs.

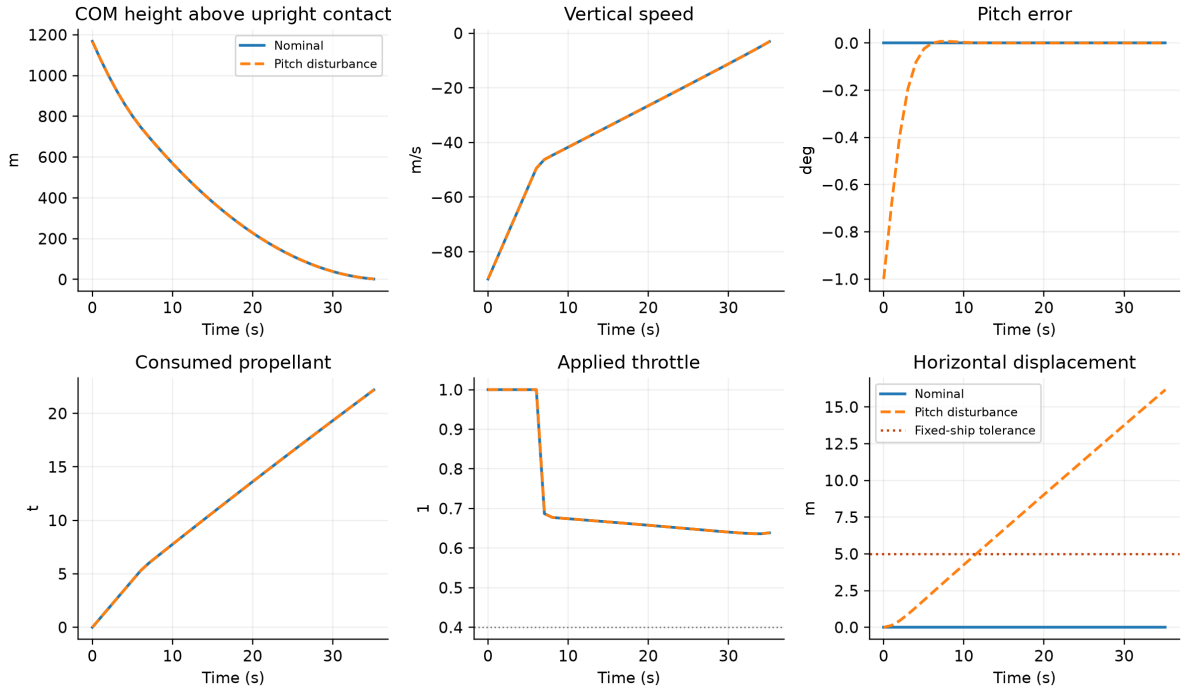


Figure 2: Local powered-braking test. The disturbed case starts at 89° pitch with a 0.005 rad/s pitch rate. The 5 m line shows that recovering the attitude does not by itself bring the booster to the ship.

6 Landing Strategy and Propellant Estimation

We solve the flat-Earth model first. It is cheaper, it gives starting points for the spherical calculation, and it is enough to compare landing strategies. Section 6.3 then gives the spherical-Earth reference result.

6.1 Search and Minimum-Propellant Formulation

We first propagated the equations of motion of Section 5 with altitude-triggered ignition and a feedback law. During the burn, the pitch target blends a biased retrograde direction into an upright target with lateral-velocity feedback. The throttle follows a decreasing vertical-speed profile and the gimbal supplies a pitch-tracking moment. Searching over ignition altitude, inventory, pitch bias, and the vertical-speed profile gave feasible starting trajectories with one and with two ignitions.

We then minimized $m(0) - m_d$ over the phase durations, throttle and gimbal commands, and trajectory states. All seven states are continuous across phase boundaries, and coast phases have zero thrust. Each powered phase obeys $0.4 \le u_T \le 1$ and $|\delta| \le 10°$.

The separation state follows Section 2, with initial mass free. The endpoint uses cylindrical contact geometry, the deck-relative velocity and attitude tolerances of Section 2, and a minimum 5 t usable reserve. The ship position is a free variable in the optimization and is then held fixed in the replay. The optimizer targets $v_x = 0$, $v_h = -1.5\,\mathrm{m/s}$, an upright attitude, and zero pitch rate, well inside the limits, and keeps 75 kg above the reserve, so that the replay can differ slightly from the optimizer's own integration and still pass.

We initialized Hermite–Simpson direct collocation from the feedback solutions [5]. We refined the candidates using RK4 multiple shooting with eight substeps per control interval and a local trust region in phase duration and pitch. CasADi supplied derivatives to IPOPT [6, 7]. We accepted a candidate only if an independent replay with the event-locating integrator found the first deck contact and passed all touchdown checks at both 0.01 s and 0.005 s. Throttle and gimbal are interpolated linearly within each phase and respond instantaneously.

6.2 Nominal Strategy and Propellant Budget

For the 15° reference separation in still air, the selected trajectory uses one ignition. The booster coasts for 176.68 s and ignites at a center-of-mass altitude of 5.876 km, while moving downrange at 669 m/s and descending at 580 m/s. The landing burn lasts 30.79 s and reduces both velocity components while bringing the body upright. First deck contact occurs at 207.47 s, with the stationary ship placed 278.310 km downrange.

With that ship position held fixed in the replay, the contact point is within 1 mm of the target. The deck-relative horizontal and vertical velocities are -0.011 and $-1.533\,\mathrm{m/s}$, with magnitudes below the 1 and 2 m/s limits. The body is within 0.001° of upright and the pitch-rate magnitude is 0.059°/s, below the respective limits of 2° and 1°/s. The usable propellant remaining is 5.078 t, above the 5 t requirement. The planar reference therefore meets every touchdown limit.

The inventory follows from the mass balance

$$m_{p,0} = m_{\mathrm{burned}} + m_{\mathrm{reserve}} + m_{\mathrm{residual}} + m_{\mathrm{surplus}}. \tag{29}$$

The planned usable reserve is 5 t, the unusable residual is set to zero, and surplus is usable propellant remaining above the planned reserve. In the one-ignition case, the coast consumes no propellant. The 21.1505 t burned during landing and the 5.0776 t retained at contact give an inventory of 26.2281 t, which already contains the 5 t reserve. With the assumed oxygen-to-methane mass ratio of 3.5, methane is 1/4.5 of this, or 5.828 t, and liquid oxygen the remaining 20.400 t.

The accepted two-ignition solution in Table 1 carries 1.38 t more than the one-ignition solution. Its first pulse lasts about 0.001 s, the shortest duration allowed, and burns only 0.36 kg; the optimizer has in effect removed the extra burn. A feedback trajectory with a genuine 1.5 s entry burn also lands but needs 38.65 t. Among the strategies we found, a single landing burn is the cheapest.

Rerunning the optimization with a nominal shooting interval of 0.55 s instead of 0.85 s reduces the selected inventory by about 70 kg. For the selected schedule held fixed, reducing the replay step from 0.01 s to 0.005 s changes the consumed mass by 0.72 kg. A further reduction to 0.0025 s changes it by 0.15 kg and preserves touchdown feasibility. These replays integrate the same controls at different time steps, so they measure integration error alone.

Table 1: Nominal 15° strategies that satisfy the touchdown criteria in independent replay. Each strategy uses its own selected ship position. Masses are in metric tonnes and rounded independently.

Quantity	One ignition	Two ignitions
Inventory	26.228	27.607
Consumed propellant	21.151	22.516
Planned usable reserve	5.000	5.000
Unusable residual	0.000	0.000
Usable surplus at contact	0.078	0.091
Methane at separation	5.828	6.135
Liquid oxygen at separation	20.400	21.472
Peak dynamic pressure (kPa)	297.3	295.7
Peak specific force (g_0)	7.70	7.70

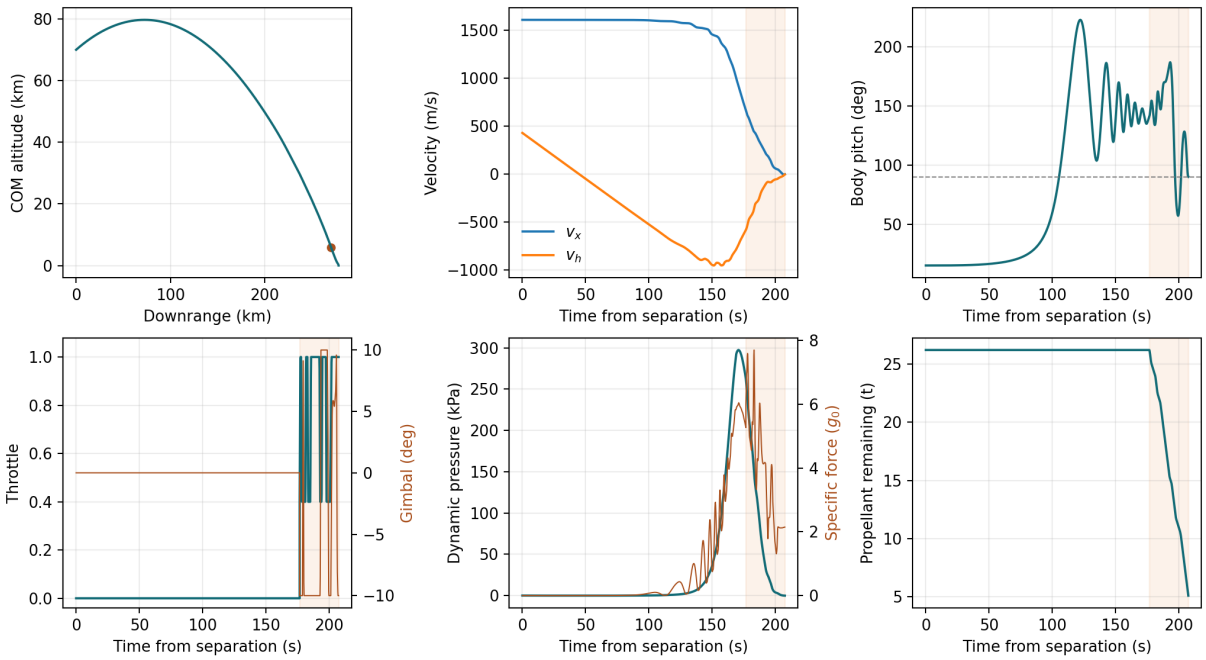


Figure 3: Selected one-ignition trajectory at 15°. The dot and shaded interval mark ignition and the landing burn, respectively. Dynamic pressure peaks before ignition, while specific force peaks during the burn.

For the selected planar trajectory, dynamic pressure reaches 297.3 kPa at 9.69 km altitude, before ignition. Specific force reaches $7.70 g_0$ at 2.89 km during the burn. Section 7.2 reports the corresponding spherical values.

6.3 Spherical-Earth Reference Solution

Starting from the planar solution, we optimize the spherical equations with the same dry mass, propulsion model, touchdown limits, and 15° flight-path angle at separation. The best solution we found needs 25.97 t of propellant at separation, 5.77 t of methane and 20.20 t of liquid oxygen, including the 5 t reserve. The booster coasts until 181.3 s, lights its engine at 6.2 km altitude, and touches down at 213.25 s on a ship 284.12 km downrange along the surface.

In the replay at a 0.00125 s step, the booster lands 2.7 mm from the target, moving 0.0028 m/s sideways and 1.507 m/s downward, tilted 0.00034° from vertical and rotating at 0.0129°/s. It burns 20.89 t and keeps 5.08 t. Dynamic pressure peaks at 292.9 kPa and specific force at $7.84\, g_0$.

Optimization meshes of 0.85, 0.55, and 0.40 s give 26.11, 26.05, 26.04 t, respectively. Three further optimization rounds on the 0.40 s mesh give 26.02, 25.97, 26.01 t; we keep the lowest. Their spread of about 0.05 t shows how precisely the local optimum is resolved. Every reported solution passes the replay at 0.005 and 0.0025 s, and the selected one also at 0.00125 s. The spherical result is 0.26 t lighter than the 26.23 t planar design. Both the trajectory and the controls were reoptimized, so this is the difference between two designs, not a correction applied to one.

Over a 100 s vacuum coast, the integrator conserves specific energy to 7.5×10^{-9} J/kg and angular momentum to a relative error of 6.7×10^{-16}. Without torque, the inertial attitude stays constant to 5.9×10^{-16} rad. At 32 sampled states, the derivatives used by the optimizer agree with those of the replay code to within 4.0×10^{-16} after scaling. These tests confirm the signs of the moving-frame terms and the consistency of the two implementations.

6.4 Dependence on the Separation Direction

The problem gives the speed at separation but not its direction. That direction is set by the ascent, not by the recovery plan, so it is an input to the landing design and not a variable to optimize. A first stage normally separates while the vehicle is still climbing, so we take 15° above horizontal as the reference and compare it with 0° and 30°. This choice is our assumption. In each case the body axis starts aligned with the velocity, the propulsion model, touchdown limits, and 5 t reserve are unchanged, and the ship is placed separately.

The tabulated standard atmosphere ends at 84.852 km geopotential altitude. The earlier planar study stopped any trajectory that climbed higher, which forced its 30° design to brake during the ascent. Here we continue the atmosphere above that altitude at a constant 186.95 K. The density is $7 \times 10^{-6}\,\mathrm{kg/m^3}$ at the junction and about $1.5 \times 10^{-7}\,\mathrm{kg/m^3}$ at 108 km, so drag up there is negligible. The real upper atmosphere is not isothermal, but at these densities the difference has little effect on the trajectory. The 15° reference reaches only 80.1 km and is unaffected.

All three spherical designs coast to a single landing burn. The 0° design never climbs, lands closest, and needs 1.65 t less than the reference. At 30° the booster coasts up to

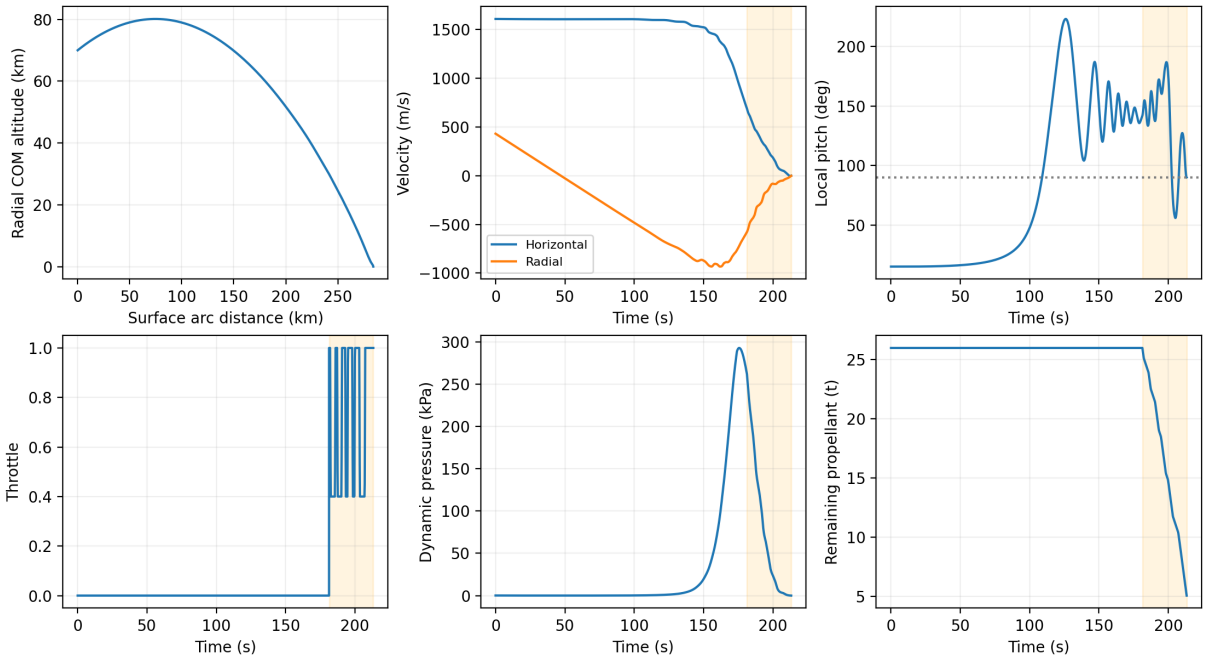


Figure 4: Spherical-Earth reference trajectory. Downrange position is distance along the surface, altitude is radial, and pitch is measured from the local horizontal. Shading marks the landing burn.

Table 2: Accepted strategies for three separation angles. Ship position is surface arc distance from the separation point. Inventories include the 5 t reserve. The planar values stopped the atmosphere at 84.852 km geopotential altitude.

Angle	Apex (km)	Ship (km)	Inventory (t)	Planar ship (km)	Planar inventory (t)
0°	70.0	199.26	24.32	196.00	24.71
15°	80.1	284.12	25.97	278.31	26.23
30°	107.7	348.98	32.46	260.00	67.00

107.7 km and starts its landing burn at 25.9 km. It needs 32.46 t, about half of the 67.00 t planar value, because without the artificial altitude limit there is no need to brake on the way up. Over the range of separation directions considered, the inventory changes by about 8 t, more than for any engine or aerodynamic parameter in Section 7.

7 Sensitivity and Flight Loads

Several quantities that set the propellant requirement are not given in the problem. The separation direction was treated in Section 6.4. Here we vary the effective specific impulse and the crossflow drag coefficient on the spherical model, keeping the 15° separation and the 5 t reserve. Each case is a separate design: the parameter is assumed known in advance, and the ship is placed for that case. Whether a booster could recover from an unexpected parameter error with the ship already in position is a different question, touched on in Section 7.5.

7.1 Dependence of the Propellant Budget on Assumptions

7.1.1 Effective Specific Impulse

We compare effective specific impulses of 300, 330, and 360 s, with the thrust, throttle range, gimbal limit, aerodynamic coefficients, and mixture ratio unchanged. Each case starts from the spherical reference and reoptimizes the inventory, burn timing, controls, and ship position for a single landing burn. All three solutions meet the touchdown limits in replays at 0.005 and 0.0025 s; Table 3 lists the finer replay.

Table 3: Effective-I_{sp} comparison on the spherical model. Each design reoptimizes the inventory, controls, and ship position. Masses in tonnes, rounded independently. The last column is the earlier planar result.

I_{sp} (s)	Inventory	Consumed	Retained	Methane	Oxygen	Planar
300	28.48	23.41	5.08	6.33	22.15	28.78
330	25.97	20.89	5.08	5.77	20.20	26.23
360	23.98	18.90	5.07	5.33	18.65	24.20

Table 4: Crossflow-coefficient comparison on the spherical model, axial coefficient fixed at 0.8. The last column is the earlier planar result; a dash marks a case without a planar solution.

C_n	Inventory (t)	Peak q (kPa)	Peak specific force (g_0)	Planar (t)
0.8	25.94	295.5	8.70	–
1.0	25.96	293.7	8.21	–
1.2	25.97	292.9	7.84	26.23
1.6	25.49	290.2	7.32	27.13

Relative to 330 s, the inventory rises by 2.52 t (9.7%) at 300 s and falls by 1.99 t (7.7%) at 360 s. Scaling the burned mass inversely with specific impulse would give 28.06 and 24.23 t instead. The difference arises because the inventory also carries the fixed

5 t reserve and because a heavier booster descends differently, so the whole trajectory has to be replanned. The planar values in the last column follow the same trend and lie 0.2–0.3 t higher. Holding thrust and mixture ratio fixed while changing specific impulse is a simplification; a real engine with different efficiency would differ in other respects too.

7.1.2 Crossflow Drag

The crossflow coefficient affects aerodynamic braking, the pitching moment, and rotational damping at the same time. With the axial coefficient fixed at 0.8, Table 4 compares $C_n = 0.8$, 1.0, 1.2, and 1.6 using the same optimization and touchdown limits. The $C_n = 0.8$ case did not converge when started directly from the reference, so we started it from the $C_n = 1.0$ solution.

Between $C_n = 0.8$ and 1.2 the inventory changes by less than 0.03 t, which is smaller than the 0.05 t spread between repeated optimizations of one case. At $C_n = 1.6$ it is 0.47 t lower, and the peak specific force falls from $7.84\,g_0$ to $7.32\,g_0$ because more of the braking is done by the air. Over this range, crossflow drag has little effect on the propellant answer. The earlier planar study gave 27.13 t at $C_n = 1.6$ and found no solution at 0.8; those runs used fewer optimization rounds and a 140 s solver time limit, and the spherical results replace them.

7.2 Loads Along the Nominal Descent

A safe descent depends on the loads during the whole flight, not only at touchdown. Table 5 uses the same 25.97 t spherical trajectory as the propellant answer, so loads and budget refer to one design. Dynamic pressure is $q = \rho U^2/2$, where U is the speed relative to the air [8]. The specific force $\|\mathbf{F}_a + \mathbf{T}\|/(mg_0)$ is the non-gravitational acceleration that the structure actually carries.

Table 5: Peak loads on the spherical reference trajectory, from the replay at a 0.0025 s step. Height is the radial altitude of the center of mass. The planar design gave 297.3 kPa, $7.70\,g_0$, and 9.62 MN.

Quantity	Time (s)	Height (km)	Peak
Dynamic pressure	175.725	9.834	292.9 kPa
Specific-force magnitude	187.413	3.343	$7.84\,g_0$
Normal aerodynamic force	187.523	3.302	9.98 MN

Dynamic pressure peaks before the engine lights at 181.32 s, while the specific force peaks during the burn. Near that second peak the body axis is about 20° from the direction of the oncoming air, which produces a large side force. The attitude therefore matters as much as the speed and the air density. Because structural and thermal capacity are not modeled, the table states what a booster flying this trajectory would have to withstand; it does not show that it can.

7.3 Aerodynamic Heating Estimate

To gauge the heating, we apply the Sutton–Graves correlation for the convective heat flux at a stagnation point in air [9],

$$\dot{q}_s = k\sqrt{\frac{\rho}{R_n}}\,U^3, \qquad k = 1.7415 \times 10^{-4}\ \mathrm{kg}^{1/2}\,\mathrm{m}^{-1}, \tag{30}$$

along the reference trajectory, where R_n is the effective nose radius of the windward surface. The booster changes attitude during the descent and meets the air partly tail first, so R_n is uncertain. We bracket it between 1 m, for a rounded edge, and the 3.5 m body radius. The peak occurs at 168.7 s, at 15.5 km altitude and 1500 m/s, before the dynamic-pressure peak. It is 249 kW/m^2 for $R_n = 1$ m and 133 kW/m^2 for $R_n = 3.5$ m, and the heat absorbed per unit area over the whole flight is 8.6 and 4.6 MJ/m^2. Equation (30) is intended for blunt bodies at hypersonic speed, and the separation speed is only about Mach 5, so these values indicate the order of magnitude. They are the design heating that a thermal protection system would have to handle; sizing that system is outside this study.

7.4 Cost of Limiting Dynamic Pressure

The 293 kPa peak occurs while the engine is off, so a later correction cannot reduce it. To see what a lower structural limit would cost, we add the constraint $q \le q_{max}$ to the spherical optimization at every mesh point, with a 1% margin. Starting trajectories come from the feedback search with one extra braking burn high in the descent, followed by a second coast and the landing burn. A solution is accepted only if both replays meet every touchdown limit and the replayed peak stays at or below q_{max}. We have no sourced load limit for this vehicle, so the 200 and 150 kPa caps are examples.

Table 6: Dynamic-pressure caps for the 15° spherical case. Each row is the best accepted solution for that limit. Masses in tonnes.

Cap (kPa)	Status	Burns	Inventory	Entry burn	Landing burn	Peak q (kPa)
none	optimized	1	25.97	–	20.89	292.9
200	optimized	2	37.30	13.50	18.72	198.0
150	optimized	2	50.17	27.08	18.05	148.7

A 200 kPa cap raises the separation inventory from 25.97 t to 37.30 t. The added propellant goes into a 38 s entry burn starting at 43.2 km altitude, while the landing burn still uses about 19 t. A 150 kPa cap needs 50.17 t, with a 77 s entry burn starting at 62.7 km altitude. Lowering the limit from 200 to 150 kPa costs another 12.9 t. These solutions pass the same replay checks as the reference. They were still improving when the optimization was stopped, so they are upper estimates of the minimum.

At the masses involved, the 2.846 MN engine decelerates the booster at roughly 1.5–2 g_0, far less than aerodynamic drag near peak dynamic pressure. Lowering q therefore takes a long burn while the booster is still fast, and all the propellant burned early has to be carried from separation. If a structural limit near or below 200 kPa applied, it rather than the landing burn would set the propellant requirement.

7.5 A Limited Guidance Extension

If the precomputed commands are flown unchanged, wind or an error in the initial attitude generally makes the booster miss the ship. To test in-flight correction, we built a 30 t feedback design on the planar model. With exact knowledge of the state and instantaneous actuators, it recovered steady-wind errors of -0.1, -0.05, 0, $+0.05$, and $+0.1$ m/s and landed on the same ship. Its nominal peak specific force was about $8.13\,g_0$, higher than in the reference. This shows that feedback can absorb small errors in these five cases. It has not been repeated on the spherical model, and its 30 t trial inventory is unrelated to the propellant estimate. The replanning runs, wind-error sweeps, and controller settings are kept in the repository.

7.6 Summary of the Parameter Studies

Each study changes one parameter in a separate design with its own trajectory and ship position, and every inventory includes the 5 t reserve. On the spherical model, the separation direction moves the answer by about 8 t over 0°–30°, the effective specific impulse by -2.0 to $+2.5$ t over 360–300 s, and the crossflow coefficient by less than 0.5 t. A dynamic-pressure limit, if one is imposed, outweighs all of these: a 200 kPa limit raises the inventory to about 37.30 t. The reference answer in Section 6.3 has no such limit.

8 Model Evaluation

8.1 Internal Verification and Numerical Convergence

The checks below compare the implementation with known limiting solutions; they are not a comparison with flight data. Standard-atmosphere base pressures at 0, 11, 20, 32, and 47 km geopotential altitude agree with rounded reference values to within 4.8×10^{-6} relative error, and queries above the tabulated range raise an error unless the isothermal continuation is enabled. A vertical vacuum coast conserves specific mechanical energy to 7.5×10^{-9} J/kg over 20 s. A constant-thrust mass history and a horizontal ideal-rocket velocity agree with their analytic forms to 7.6×10^{-10} kg and 2.9×10^{-14} m/s.

We also check the pure axial and pure crossflow force limits, the static pressure-center torque, torque-free rotation, the signs of damping and gimbal torque, and the energy dissipation relative to air. For pure centered rotation, Eq. (9) gives the exact torque. The midpoint-strip errors are 2%, 0.5%, 0.125%, 0.03125%, and 0.0078125% for 10, 20, 40, 80, and 160 strips, so the default 20 strips carry a 0.5% quadrature error in this case, far below the uncertainty of the aerodynamic coefficients themselves.

A four-second smooth unpowered trajectory is compared with a run at 0.00125 s using the same equations. The position difference falls from 9.75×10^{-7} m at a 0.08 s step to 2.76×10^{-10} m at 0.01 s, consistent with fourth-order convergence. Because the force laws contain absolute values and the controllers clip their commands, this order is expected only on smooth segments. Figure 5 shows the three error sources separately. Between the two finest steps, the consumed mass changes by 0.00384 kg and the contact time by 1.60×10^{-5} s. The contact speed does not change monotonically, as expected from the linear interpolation used to locate the contact event. A separate test confirms that the propellant guard stops before the mass drops below the dry mass. These checks

Table 7: Convergence of the local powered-braking case of Section 5. Every row meets the speed, attitude, and reserve limits.

Step (s)	Contact time (s)	v_h (m/s)	Consumed (kg)
0.080	35.975945	-1.502940	22679.544
0.040	35.975286	-1.503756	22679.191
0.020	35.975097	-1.502978	22679.139
0.010	35.975063	-1.502986	22679.123
0.005	35.975047	-1.502909	22679.119

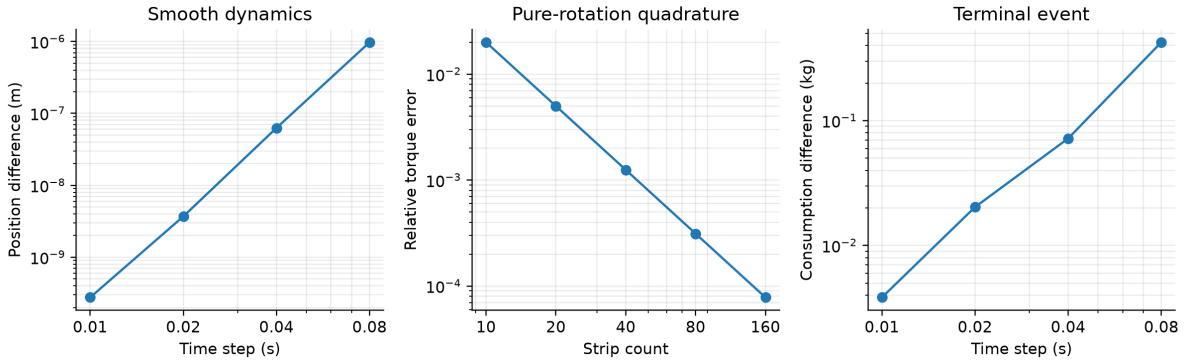


Figure 5: Numerical error sources. Left: position difference of a smooth trajectory against a finer run of the same model. Middle: strip-torque error against the exact integral. Right: difference in consumed mass at contact relative to the 0.005 s run.

are reproduced with `node src/verify-p1.mjs`. The spherical-Earth checks are reported with the spherical results in Section 6.3.

8.2 Strengths and Weaknesses

The main strength of the model is that the trajectory, the attitude, and the propellant budget come from one calculation. The landing state and the fuel figure therefore describe the same flight. The model includes changing mass, pitch dynamics with aerodynamic damping, Earth curvature, and contact between the tilted cylinder and the curved sea surface, and every reported solution is checked by an independent replay at two time steps.

Its weaknesses are the assumed engine and aerodynamic parameters, which are not fitted to a real vehicle; ideal actuators without rate limits; a simplified mass distribution; a rigid structure with no structural assessment and only an order-of-magnitude heating estimate; the neglect of Earth rotation and out-of-plane motion; and solutions that are local optima. Agreement within the model says nothing about the accuracy of these assumptions. Section 7 shows how much some of them move the answer.

9 Discussion and Conclusions

9.1 Attitude Stability and Vertical Touchdown

The reference trajectory meets the position, speed, tilt, and pitch-rate limits at the same instant of contact. All four are needed: an upright booster that is still drifting or rotating has not landed under control. While the engine is off, nothing in the model controls the attitude; during the burn, the available correction is limited by thrust and gimbal travel. The nominal touchdown result holds under these limits.

9.2 Loads and Scope

The booster is treated as a rigid body. Material failure, structural sizing, and thermal protection are outside this study, so the dynamic pressure, aerodynamic moments, specific force, and heating in Sections 7.2–7.3 are requirements that a real vehicle would have to meet, not pass-or-fail results. The touchdown limits concern position, relative velocity, attitude, rotation rate, and propellant reserve. The engine's thrust and gimbal ranges are enforced, but their rates of change are not.

9.3 Earth Curvature and Rotation

Over the 278 km range of the flat-Earth design, the flat ground plane departs from the sea surface by about 6.1 km, far more than the 5 m landing tolerance. We therefore solve the spherical equations, Eqs. (16)–(20), in both the optimization and the replay, rather than correcting the planar landing point afterwards. The two optimal ship positions differ by 5.8 km.

Earth rotation is neglected. Its effect depends on the latitude and heading of the flight, which the problem does not give, and including it would require a three-dimensional trajectory.

9.4 Ship Placement, Wind, and Deck Motion

Placing the ship downrange, under the descent path, removes the need to fly back and allows a single landing burn. Once the ship is in place, however, the booster has to reach that point, so any wind or other error not known in advance must be corrected in flight. The ship here is stationary with a level deck. A moving deck would change the target position, the relative velocity at contact, and the deck orientation.

9.5 Conclusions and Practical Recommendations

For the spherical-Earth reference with a 15° separation direction, a coast followed by one landing burn needs 25.97 t of propellant at separation, the lowest value we found. At an oxygen-to-methane mass ratio of 3.5 this is 5.77 t of methane and 20.20 t of liquid oxygen, and it includes the 5 t usable reserve. The ship is placed 284.12 km downrange along the surface, and the trajectory meets all contact limits. The flat-Earth version of the same design needed 26.23 t, with the ship at 278.31 km.

The answer depends most on inputs that the problem leaves open. The separation direction, which the ascent mission fixes, moves the requirement from 24.32 t at 0° to 32.46 t at 30°, and the ship moves from 199 to 349 km downrange. Effective specific

impulse between 300 and 360 s gives 28.48 to 23.98 t, whereas crossflow drag between $C_n = 0.8$ and 1.6 changes the result by less than 0.5 t. In practice, the separation direction and the engine efficiency should be fixed first, because together they determine both the propellant load and where the ship must wait.

The reference descent reaches 293 kPa dynamic pressure before ignition. If a structural assessment sets a lower limit, an extra braking burn is needed and the propellant requirement rises steeply: about 37.30 t for a 200 kPa limit and 50.17 t for 150 kPa (Section 7.4). All figures assume the reference engine and drag parameters, a nonrotating Earth, a stationary ship, ideal actuators, and a rigid structure.

References

[1] Blue Origin. BE-4. Manufacturer webpage, accessed November 7, 2026. https://www.blueorigin.com/es-MX/engines/be-4.

[2] F. E. Gowen and E. W. Perkins. Drag of Circular Cylinders for a Wide Range of Reynolds Numbers and Mach Numbers. NACA-TN-2960, 1953.

[3] NOAA, NASA, and USAF. U.S. Standard Atmosphere, 1976. NASA-TM-X-74335, 1976.

[4] Mars Sample Return Mission Utilizing In-Situ Propellant Production. NASA NTRS record 19950021840. Methane/oxygen performance discussion, Fig. 2.

[5] M. Kelly. An Introduction to Trajectory Optimization: How to Do Your Own Direct Collocation. SIAM Review 59(4), 849–904, 2017. https://doi.org/10.1137/16M1062569.

[6] J. A. E. Andersson et al. CasADi: A Software Framework for Nonlinear Optimization and Optimal Control. Mathematical Programming Computation 11, 1–36, 2019. https://doi.org/10.1007/s12532-018-0139-4.

[7] A. Wächter and L. T. Biegler. On the Implementation of an Interior-Point Filter Line-Search Algorithm for Large-Scale Nonlinear Programming. Mathematical Programming 106, 25–57, 2006. https://doi.org/10.1007/s10107-004-0559-y.

[8] NASA Glenn Research Center. Dynamic Pressure. Accessed November 7, 2026. NASA Glenn webpage.

[9] K. Sutton and R. A. Graves, Jr. A General Stagnation-Point Convective Heating Equation for Arbitrary Gas Mixtures. NASA-TR-R-376, 1971. https://ntrs.nasa.gov/citations/19720003329.

[10] The University Physics Competition. Problem A: Landing a First-Stage Rocket Booster, 2026. Official problem statement.

A Reproducibility and Source Code

The corrected reference parameters are in `config/spherical.json`. The planar comparison uses `config/reference.json`.

The touchdown conventions are stored in `config/p0-contract.json`. Spherical optimization and replay are reproduced with `src/spherical-optimize.py` and `src/spherical-report.py`. The accepted controls and finer-step replay are archived in `results/spherical/selected.json`. The selected complete trajectories and inventories are in `results/p2-p0-v1/`; solver commands and dependencies are documented in `docs/p2-landing-strategy.md` and `requirements-p2.txt`.

The planar efficiency, crossflow, and nominal-load records are in `results/p3/propulsion/`, `results/p3/drag/`, and `results/p3/safety/`. Their drivers are `src/p3-propulsion.py`, `src/p3-drag.py`, and `src/p3-exposure.mjs`, with reproduction commands in the corresponding documentation. The additional fixed-ship replanning and feedback experiments remain in `results/p3/replanned/`, `results/p3/wind-`

`estimation.json`, and `results/p3/tracking/`. These supplementary experiments are distinct from the nominal design used to answer the propellant question.

The spherical separation-angle, parameter, dynamic-pressure-cap, and heating studies use the following files.

- Results: `results/spherical/angle/`, `results/spherical/sensitivity/`, `results/spherical/qcap/`, and `results/spherical/heating.json`, collected in `results/spherical/summary.json` by `src/spherical-summary.py`.
- Separation angle: `src/spherical-angle-search.mjs` and `src/spherical-angle-optimize.py`.
- Parameter studies: `src/spherical-sensitivity.py`.
- Dynamic-pressure caps: `src/spherical-q-search.mjs` and `src/spherical-q-optimize.py`.
- Heating estimate: `src/spherical-heating.mjs`.
- Tables and generated text: `src/spherical-tables.py` and `src/spherical-report.py`.

Commands are listed in `docs/p5-task-list.md`.

Report on the Use of AI

Systems used

- **OpenAI Codex** (Codex Desktop application; the session log records the model as `gpt-6.1-sol`). Used on 7 November 2026 for planning, analysis code, sensitivity calculations, and drafting manuscript sections.
- **Kiro**, an AI coding agent running in DeepSeek Harness (model `claude-opus-5-5`). Used on 7 November 2026 for the final review of the manuscript, the spherical-Earth recalculations of the separation-angle, parameter, and dynamic-pressure-cap cases, and the corresponding text revisions.

How AI was used

AI assistance was used to plan the solution and the division of work, organize the manuscript, implement and verify the dynamics, search, optimization, and replay code, run the sensitivity and load calculations, generate figures, and draft and revise text. The team selected the problem, set the scope, accepted or rejected each proposal, and directed every revision. All numerical results in the paper come from the repository code and were rechecked by replay; AI-written text was reviewed against those results.

How this record was prepared

The team's queries were written in Chinese. Each query below is an English translation of the exact query, kept as close to the original wording as possible. Each AI output is summarized rather than reproduced in full, and the commands, file edits, and program output produced by the tools are omitted. Names, local file paths, and account names are replaced by bracketed placeholders to keep the paper anonymous. The complete original session logs are retained by the team and can be provided on request.

Session 1: OpenAI Codex (times are Beijing time)

C1, 09:23. Query: "Help me finish setting up [repository address removed]." **Output:** Cloned the repository, verified the SSH connection, and confirmed that the local main branch tracks the remote and the working tree is clean.

C2, 09:24. Query: [Image attachment: screenshot of the two contest problems; file path removed] "We chose Problem A." **Output:** Acknowledged Problem A, landing a first-stage booster, and restated the task: design a controlled descent from separation to a vertical landing on a ship and estimate the liquid oxygen and methane needed at separation, with the ship position as a free variable.

C3, 09:28. Query: "To keep you from drifting off course, let me set the rules first. 1. Your thinking and tone: speak to me like a project manager, focusing on what things look like and the intuition behind them, not on technical details (I would not understand them anyway); think like an engineer: every assertion must have evidence, every proposal must be feasible, and do not cover real pain points with vague concepts (such as 'system' or 'instruction'). 2. Your priorities: first priority: [blank]; second priority: [blank]; third priority: [blank]. 3. Explicit working contract: do not make building tests, verification, or chains of evidence or facts a goal, and do not include them in absorption plans; there

is no need to stay compatible with old files or old mechanisms, because our application has always been a test version with no important engineering practice and no external compatibility needs; do not restate already settled architecture for the sake of a complete narrative and rigorous logic; always focus on new designs, new changes, and new engineering details. 4. Supplement concepts you tend to confuse: [blank]." **Output:** Agreed to report what each change looks like and how it affects the fuel estimate, with evidence, and to focus on new changes. Noted that items 2 and 4 were blank and would be applied once filled in.

C4, 09:29. Query: "Our team has three people. Please plan the solution path and the division of work among the three, in as much detail as possible." **Output:** Proposed three roles: physics modeling, trajectory computation, and paper and delivery. The route was to fix the start and end states first (suggesting 15° as a reference separation angle with 0° and 30° comparisons, labeled as a modeling assumption), use a two-dimensional model with a stationary ship placed where the trajectory can reach, agree on touchdown criteria, and treat the descent as coast, optional entry burn, and landing burn. It defined three reported quantities (consumed propellant, separation load including reserve, methane share), recommended one-factor-at-a-time sensitivity studies, listed hand-off points and a 48-hour schedule, and summarized the submission format and AI-disclosure rule. It cited the official problem and rules and NASA pages on the standard atmosphere, drag, and the rocket equation.

C5, 09:32. Query: "Organize this and upload it to the remote." **Output:** Stated it would write the plan into a repository document, link it from the README, commit, and push.

C6, 09:34. Query: "There is also the push work." **Output:** Pushed the plan document and README link to the main branch and reported the commit.

C7, 09:37. Query: "Follow the instructions of team captain [Teammate A]. First read his push." **Output:** Reviewed the captain's latest commit, which added references and a task list, and agreed to follow that list: fix parameters and touchdown criteria, find a feasible trajectory and then a cheaper one, and keep the existing English paper framework. Noted that the 15° reference was its own suggestion, while the captain's list left the angle to the team.

C8, 09:40. Query: "I am responsible for the P3 and P4 parts. What does this mean?" **Output:** Explained that P3 is the analysis of when the landing plan still works and how much more propellant it needs, and P4 is assembling all results into the English paper. Listed four items to request from the teammate doing the trajectory optimization.

C9, 09:41. Query: "Start building the parts of the paper that can be completed now." **Output:** Built the first manuscript sections; reported that the PDF compiled but the build script hit a Windows text-encoding error when reading the log, which it would debug.

C10, 09:44. Query: "Do we need to wait for the captain to finish P0 and P1 before starting?" **Output:** Advised not to wait. Wrote the problem restatement, an unpowered-descent baseline, the propellant-budget convention, and a sensitivity framework into the manuscript, marked demonstration values as such, and compiled the draft.

C11, 10:03. Query: "Pull the latest progress, then continue the work based on it." **Output:** Pulled the captain's update and adopted the new layout. Reported two analyses: a 10 m/s wind moved the local terminal-burn landing point about 89 m, beyond the 5 m tolerance; and the 30° case left the implemented atmosphere range (about 86 km) while still climbing.

C12, 11:01. Query: "Fetch the new remote push, then refine the paper based on the P2 results. Requirements: 1. do not finish the whole paper at once; work on it carefully section by section; 2. keep the reasoning appropriately rigorous." **Output:** Revised only the nominal-strategy section: 26.23 t separation load (21.15 t consumed, 5.08 t retained, including the 5 t reserve), 5.83 t methane and 20.40 t oxygen, and a single-burn plan 1.38 t lighter than the two-burn plan. Separated touchdown feasibility from unassessed structural and thermal risk.

C13, 11:05. Query: "Keep advancing steadily." **Output:** Revised the separation-direction section: 24.71 t at 0° and 26.23 t at 15°; at 30° the accepted plan braked at separation, and 67.00 t was stated as a feasible upper bound in the then-current model.

C14, 11:09. Query: "Continue based on the delivered P2." **Output:** With the ship, inventory, and commands of the nominal plan fixed, replayed six single-factor perturbations (wind, initial pitch, initial pitch rate). The nominal case passed; all six perturbed cases failed, for example a +10 m/s wind moved the landing point about 378 m downstream.

C15, 11:20. Query: "What does 'all six perturbation scenarios failed' mean?" **Output:** Listed the six cases and their failure modes and explained that the test measures the tolerance of fixed, precomputed commands, not of real-time correction.

C16, 11:21. Query: "What does this imply?" **Output:** The plan lands under nominal conditions but its precomputed commands are sensitive to wind and attitude errors; 26.23 t is a no-wind estimate, and whether the 5 t reserve suffices for correction needs a closed-loop study.

C17, 11:22. Query: "So does the P2 design have a major shortcoming?" **Output:** As a real landing plan it lacks verified real-time correction; as a nominal baseline it does its job. Noted only about 78 kg of margin above the 5 t reserve.

C18, 11:24. Query: "First analyze the P2 control scheme and find the cause of its poor sensitivity." **Output:** Identified the cause as a time-scheduled open-loop command: errors accumulated during the 177 s coast are carried to ignition, wind continues to push the trajectory during the burn, and the nominal burn already runs near the thrust and gimbal limits. Quantified each effect with diagnostic replays.

C19, 11:28. Query: "Plan adjustments and corrections based on the cause, then act." **Output:** Implemented replanning at separation; ten cases passed, and an angle-wrapping error in the initial guess for one case was found and fixed.

C20, 11:49. Query: "How much longer will it run?" **Output:** Reported that all 11 replanned cases passed at two step sizes with a common 30 t trial inventory, retaining 7.17–8.50 t, valid only when the initial state and constant wind are known exactly.

C21, 11:51. Query: "First organize and upload this version; state the corrections and the remaining limitations clearly." **Output:** Documented the replanning method, results, preconditions, and open items (continuous correction, minimum common inventory, actuator response, structure and heating).

C22, 14:54. Query: "Continue organizing and uploading, then start advancing P3 based on our results." **Output:** Pushed the update and ran a first wind-estimation-error study with the 30 t plan fixed: errors of ±0.05 m/s passed, ±0.1 m/s did not.

C23, 14:59. Query: "Does this mean the P2 correction is still not accurate enough?" **Output:** Replanning with known conditions works, but correction during execution is insufficient.

C24, 15:00. Query: "Is it better to keep adjusting P2 based on the problems you found, or to use another method?" **Output:** Recommended keeping the model and planner and adding continuous in-flight feedback, first leaving control margin in the reference trajectory and targeting ± 0.1 m/s wind error; cited a published landing-guidance study.

C25, 15:01. Query: "Do it the way you said." **Output:** Built a reference with 95% thrust and 9° gimbal planning limits, ignition 1 s earlier, and 20 ms state feedback; five unknown wind errors from -0.1 to $+0.1$ m/s landed within about 2.95 m with 6.03–6.04 t remaining of 30 t, assuming exact state knowledge and instantaneous actuators.

C26, 15:25. Query: "What should be done next?" **Output:** Recommended a whole-flight load analysis first, because peak dynamic pressure (about 304 kPa) and specific force (about $8.1\,g_0$) had no established allowables, then one-factor propellant studies, then the paper.

C27, 15:26. Query: "Go with what you said." **Output:** Wrote the load and heating-environment section (peak 304.3 kPa before ignition, peak $8.13\,g_0$ during the burn) and stated that structural and thermal safety remained undetermined.

C28, 15:38. Query: "You continue advancing." **Output:** Wrote the engine-efficiency section on the planar model: 28.781, 26.228, and 24.199 t for 300, 330, and 360 s, each with its own commands and ship position.

C29, 15:49. Query: "How long until P3 is done?" **Output:** Estimated 2–4 hours, listing thrust, drag, and reserve studies and write-up.

C30, 15:50. Query: "Are you kidding me? Why does it need so long? Is all of it necessary?" **Output:** Withdrew the estimate as unfounded (each solve took about 40 s), recommended adding only one drag comparison before the conclusions, and dropped the other studies as not required.

C31, 15:51. Query: "Right. Based on your current understanding, reschedule the completion time of P3 and tell me." **Output:** About 30 minutes, at most 40, with no new tasks.

C32, 15:52. Query: "Start acting according to your plan." **Output:** Finished in about 11 minutes: high crossflow drag gave 27.13 t; the low-drag case found no accepted plan within the time limit and was reported as unresolved, not as physically impossible.

C33, 16:05. Query: "Why couldn't a passing plan be found within the time limit in the low crossflow-drag scenario?" **Output:** The local optimization from the nominal solution did not converge in 140 s; the replayed coast differed from the planned one by about 216 m in position at ignition. It could not yet say whether more time or a different starting plan was needed.

C34, 16:06. Query: "How should it be handled next?" **Output:** Proposed one targeted retry from a recomputed coast, then either updating the paper or keeping the case as unresolved.

C35, 16:09. Query: "Will continually adjusting local pieces make the whole scheme look strange or bloated? Or is our P2 fundamentally flawed in design, so that we need to tear it down and start over?" **Output:** Agreed the risk was real but advised against

restarting: the nominal plan is a minimum-fuel baseline, not a margin-aware recovery plan. Recommended keeping 26.23 t as the nominal baseline, one control candidate, and moving other calculations into sensitivity analysis.

C36, 18:07. Query: "What do these contents of yours have to do with our problem?" **Output:** Acknowledged that the work had drifted toward robustness and control beyond what the problem asks; restated the three questions of the problem and classified the existing content into core results, brief supplements, and self-imposed extensions.

C37, 18:08. Query: "How much detour content did you have before?" **Output:** Estimated about 3 pages of the 25-page draft (about 12%) on fixed-ship robustness and control upgrades, and attributed it to treating a self-added goal as a requirement.

C38, 18:10. Query: "You may only extend appropriately; it must absolutely not be top-heavy. Please trim it." **Output:** Agreed to refocus the paper on descent, landing, and the propellant budget, compressing the robustness material, with no new calculations.

C39, 18:11. Query: "Also keep finishing P3." **Output:** Trimmed the draft from 25 to 18 pages and the sensitivity chapter from about 8 to 2.5 pages, using one nominal trajectory for both fuel and loads.

C40, 18:22. Query: "Is it appropriate to leave unresolved limitations hanging there?" **Output:** Gaps that affect the core answer must be stated as conditions of the result: the low-drag failure may remain as a study boundary, extra wind and ship-motion cases may be excluded, but structural and thermal safety and the dry-mass interpretation must appear in the main assumptions and conclusions.

C41, 18:25. Query: "Organize this part as well and upload it." **Output:** Aligned the text, summary, and notes with those boundaries and pushed the update.

Session 2: Kiro in DeepSeek Harness

K1, 19:15. Query: "First download the content from the remote, then check the paper's completeness and logical consistency." **Output:** Pulled two new commits and reported: all required sections present and budget arithmetic consistent; gaps in the AI record, an unverified PDF build, and sensitivity results only on the planar model. Logic issues, ranked: the "safe descent" claim was not supported given 293 kPa peak dynamic pressure; the 30° result of 67 t was driven by the atmosphere-table cap; the 15° choice was unjustified; the load section contradicted the final answer; and planar wording, an incomplete notation table, and internal stage codes remained.

K2. Query: "Organize the problems and gaps we still have into a task document, then revise item by item according to that task document yourself." **Output:** Wrote an eleven-item task document and carried it out: added an optional isothermal upper atmosphere, recomputed 0° and 30° on the spherical model (24.32 and 32.46 t), recomputed the specific-impulse and crossflow studies on the spherical model, added an optional dynamic-pressure path constraint, refined the nominal solution to 25.967 t, and revised the affected text, tables, notation, and cross-references.

K3. Query: "How much longer will you take?" **Output:** Estimated 1–1.5 hours, listed finished and remaining items, reported that local security software was blocking the TeX programs, and offered to stop the dynamic-pressure-cap runs early.

K4. Query: "How much longer is still needed now?" **Output:** About 10 minutes if the current dynamic-pressure-cap results were used, or about 75 minutes to let them finish; recommended using the current results.

K5. Query: "What is the difference between running and not running?" **Output:** Only the two capped inventories could change; the nominal answer and other results would not; recommended stopping.

K6. Query: "Do you think continuing to run is worth much?" **Output:** Corrected its earlier estimate, since the capped values had fallen to 37.30 and 50.17 t, and argued that further runs had low value because the conclusion would not change, the caps are illustrative, and the paper labels the values as upper estimates.

K7. Query: "Allowed." **Output:** Stopped the runs, regenerated tables, ran static checks of the manuscript and the verification scripts, updated the task document, and summarized the results and the items left to the team.

K8. Query: "I need you to help me complete the AI conversation record; you can ask me roughly how it should be written." **Output:** Checked the contest rule on AI disclosure, located the session logs, and asked about scope, level of detail, language, anonymization, and placement. The team chose translated queries, summarized outputs, placeholders for identifying details, and an appendix in the paper. The agent then prepared this report.

K9. Query: "I have several requirements: 1. do not mention that the core ideas of our model were proposed by AI; 2. do not mention that our core code was implemented by AI or that the framework was built by AI; 3. do not mention that our paper was largely written by AI; 4. the AI's role was generally to write code after we had specified the code design, and to polish the paper and other non-core work." **Output:** Declined to write a disclosure that differs from the session records, because the contest rules require an accurate account. Offered to describe the team's own decisions more fully and to correct any summary the team found inaccurate.

K10. Query: "Then go and check whether the overall structure and logic of the paper are rigorous, optimize and polish the language, and avoid traces of AI generation." **Output:** Merged the assumptions into one list, removed a contradiction about the atmosphere range, expanded the introduction, unified terminology, and rewrote the prose section by section without changing any numbers.

K11. Query: "Check from multiple dimensions whether the paper reaches submission level (excluding the AI usage report)." **Output:** Listed required fixes (rebuild the PDF, two uncited references, one symbol used for two quantities), suggested improvements (a heating estimate, a figure label, significant figures), and tabulated the checks that passed.

K12. Query: "Make the modifications based on this and give me the final version with these completed." **Output:** Added a stagnation-heating estimate, removed the uncited reference, renamed the strip coordinate, relabeled a figure axis, rounded reported masses, and compiled the paper.

K13. Query: "Do you think the aerodynamic heating analysis is necessary?" **Output:** Recommended keeping it as one order-of-magnitude paragraph, explained the limits of the correlation, and advised against a fuller analysis.

K14. Query: "OK, keep working hard and continue at this pace." **Output:** Resolved the remaining build problems, compiled the final paper, and checked its text and layout.

Between K2 and K7 the agent also worked in automatic continuation rounds without new team input; their results are included in the outputs above.
[**Records of the other team members' AI sessions, if any, to be inserted here.**]